

## Shifting 2D lines

### Note:

- Here we are working in the co-ordinate geometry space and not working with vectors

### Q. How to shift line by 'a' units to the right?

- First let's ensure that the line passes through the origin for that we'll make the bias term equal to 0

### Points to remember:

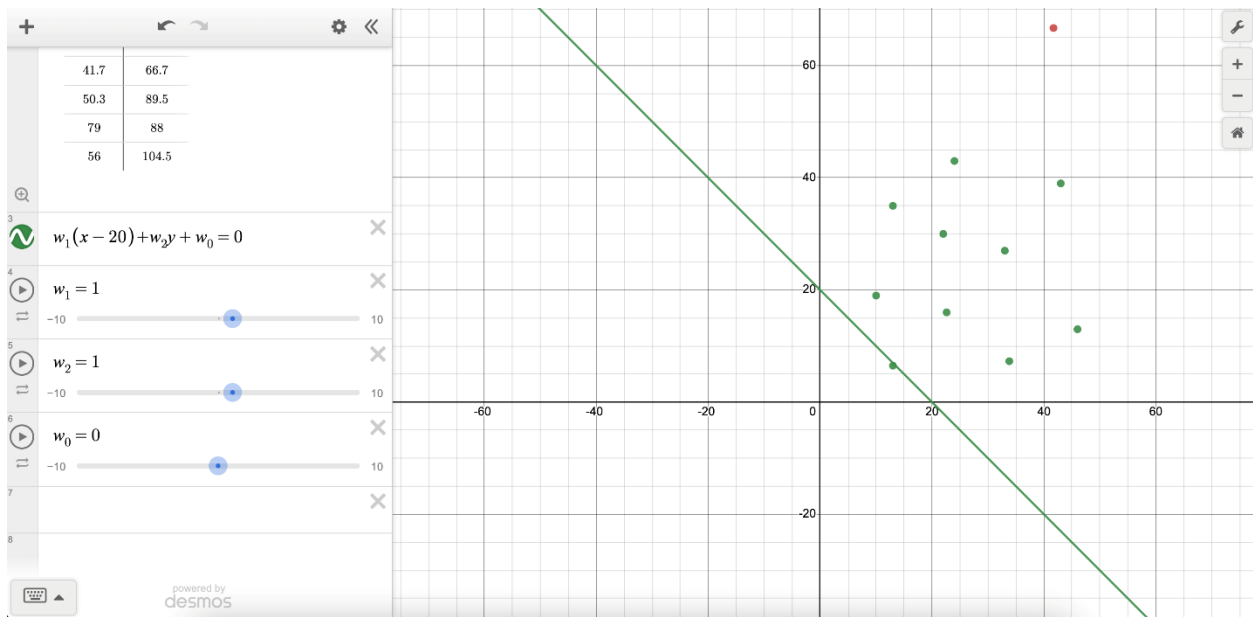
**Note:** Here origin moving means position of origin with respect to the line

- When shifting a line towards the right the origin moves towards left and if we shift the line to the left then the origin is moving to the right.
- When shifting the line up our origin is moving down and when shifting line down our origin is moving up
- Thus we can say that the origin is moving in the opposite direction to where we shift our line

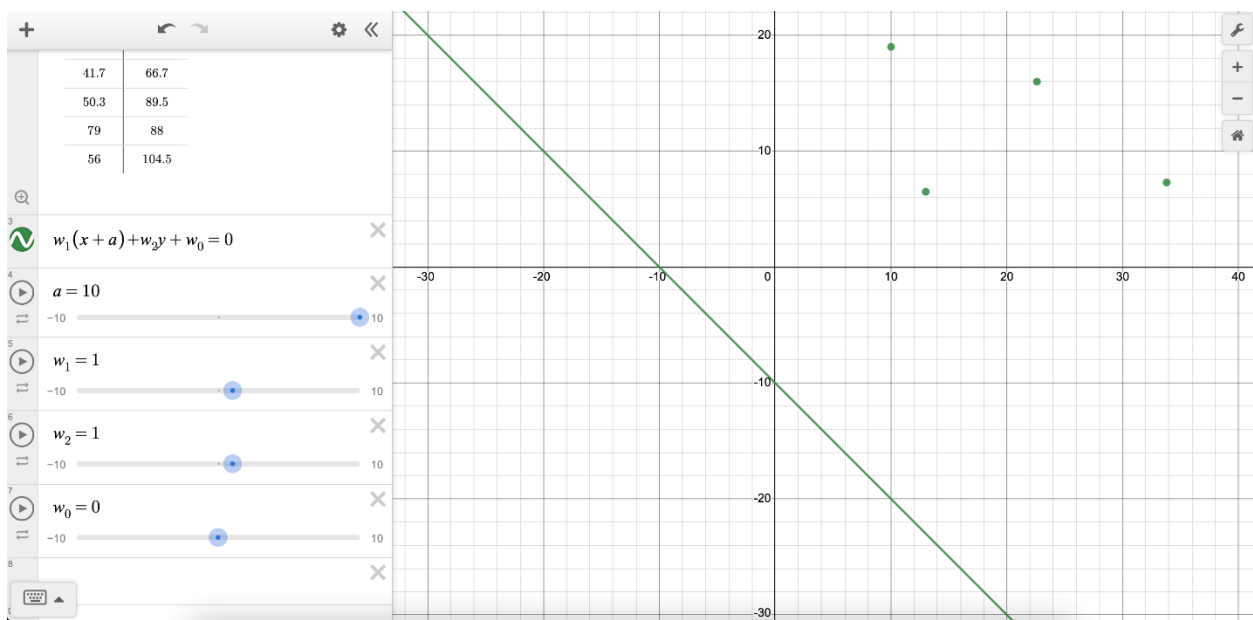
### Let's see how to represent this mathematically

**Q. When we are asked to shift a line to the right what will we change in the equation of line  $w_1x_1 + w_2x_2 = w_0 = 0$ ?**

- Since the line is passing through origin and we want it to move towards the right left say by 20
- we also know that when we move the line towards right the origin should be on the left side of the line
- so what we'll be doing is we'll be moving origin by 20 units to the left which ensures that the origin is towards the left hand side of the line or we can also say that we are moving the line towards right.
- thus the updated equation of the line is  $w_1(x_1 - 20) + w_2x_2 = w_0 = 0$
- **Always think of it as moving the origin and you'll not get confused.**
- So whenever we say that we have shifted the line by 10 units to the right essentially we can say that we have shifted the origin towards left by 10 units.



Desmos Link: <https://www.desmos.com/calculator/l4n0qwejjl>



We can use the same logic to move the line up and down by a units

## Shifting 2D Lines

$\alpha =$

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

How to shift a line by “a” units to the right?

$$w_1 (x_1 - a) + w_2 x_2 + w_0 = 0$$

How to shift a line by “a” units to the left?

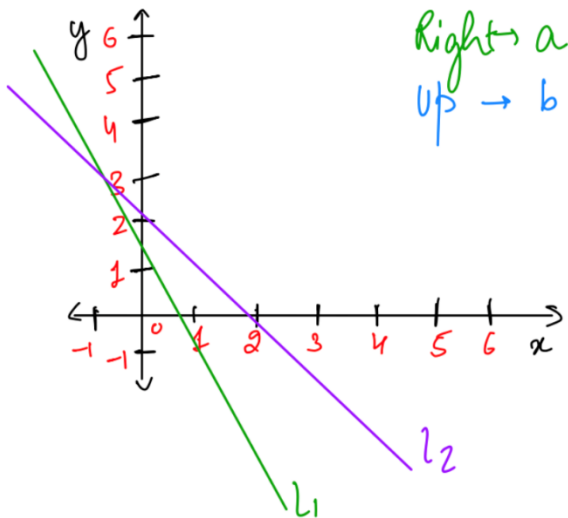
$$w_1 (x_1 + a) + w_2 x_2 + w_0 = 0$$

How to shift a line by “a” units up?

$$w_1 x_1 + w_2 (x_2 - a) + w_0 = 0$$

How to shift a line by “a” units down?

$$w_1 x_1 + w_2 (x_2 + a) + w_0 = 0$$



$$w_1 (x_1 - a) + w_2 (x_2 - b) + w_0 = 0$$

$$w_1 x_1 + w_2 x_2 - \underbrace{w_1 a - w_2 b + w_0}_{w'_0} = 0$$

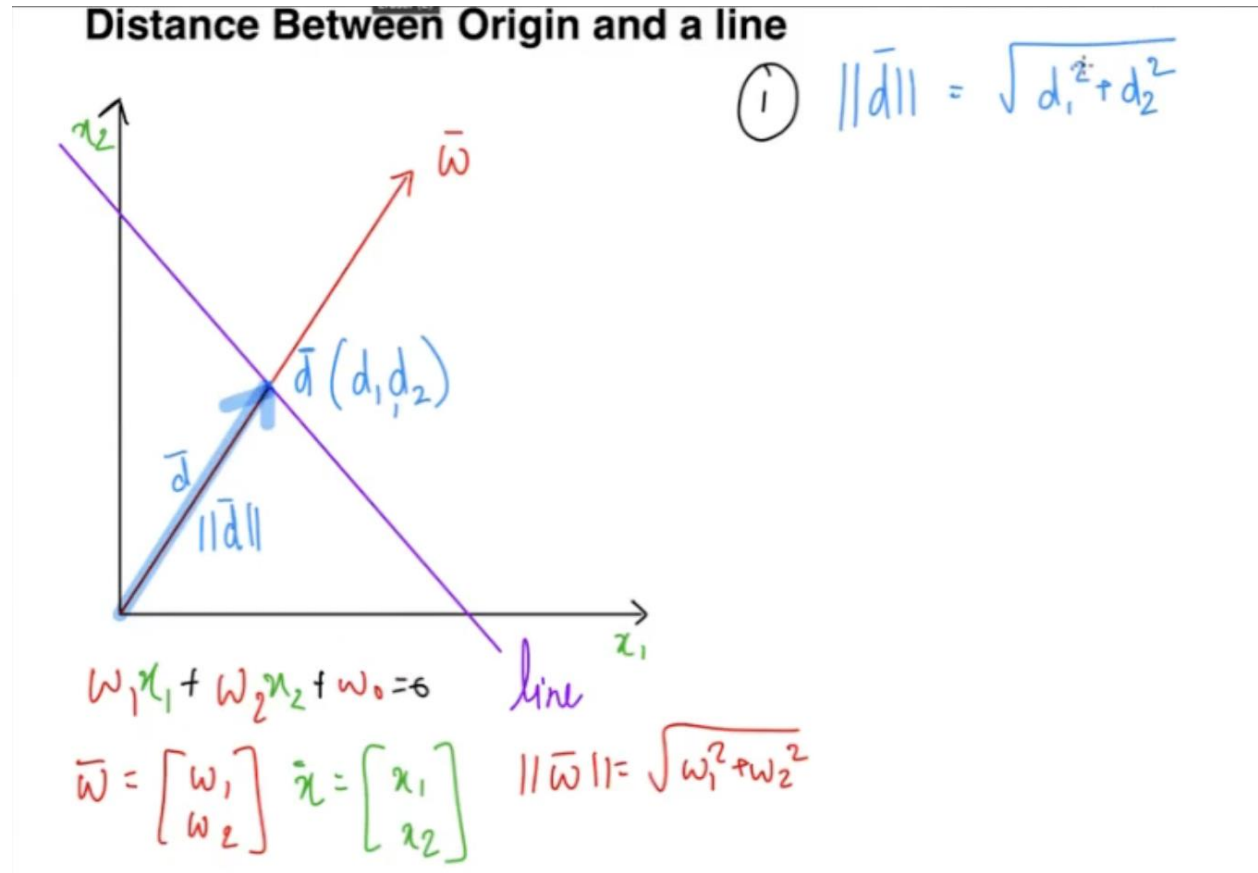
$$w_1 x_1 + w_2 x_2 + w'_0 = 0$$

$$w'_0 = -w_1 a - w_2 b + w_0$$

Let's understand why we learnt all the these

Proof of Weight vector is always perpendicular to the Line or any hyperplane

In the last class we said that the weight vector and the classifier(Line) are always perpendicular to each other now we want to prove this



**Q. Can we say that the point d is lying on the line L1?**

- Yes

**Q. which also means that it should satisfy the equation of the line L1?**

- Yes it should

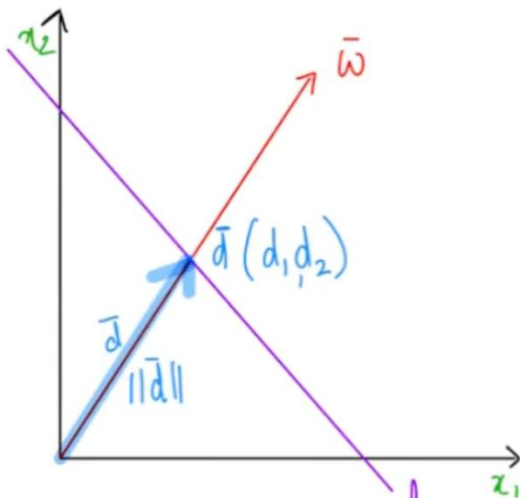
**Q. Thus can we write the equation of line as  $w_1 d_1 + w_2 d_2 + w_0 = 0$ ?**

- Yes

**Q. Can we say that the d vector is some random constant 'k' times the weight vector w?**

- Yes

## Distance Between Origin and a line



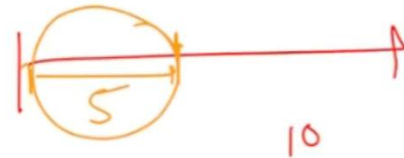
$$w_1 x_1 + w_2 x_2 + w_0 = 0 \quad \text{line}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \|\bar{w}\| = \sqrt{w_1^2 + w_2^2}$$

$$(1) \quad \|\bar{d}\| = \sqrt{d_1^2 + d_2^2}$$

$$(2) \quad w_1 d_1 + w_2 d_2 + w_0 = 0$$

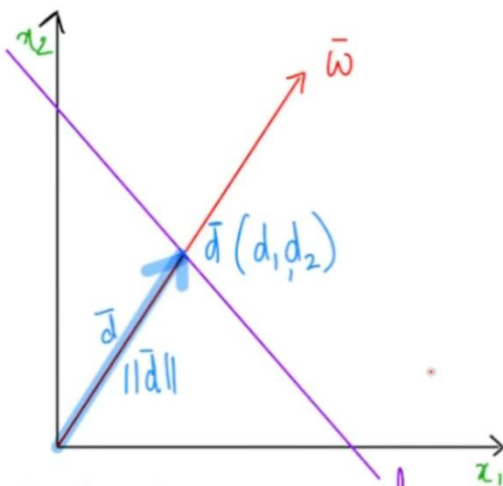
$$(3) \quad \hat{w} = \frac{\bar{w}}{\|\bar{w}\|} \quad \bar{d} = k \hat{w}$$



$$5 = \left(\frac{1}{2}\right) \times 10$$

$$d = k \bar{w}$$

## Distance Between Origin and a line



$$w_1 x_1 + w_2 x_2 + w_0 = 0 \quad \text{line}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \|\bar{w}\| = \sqrt{w_1^2 + w_2^2}$$

$$(1) \quad \|\bar{d}\| = \sqrt{d_1^2 + d_2^2}$$

$$(2) \quad w_1 d_1 + w_2 d_2 + w_0 = 0$$

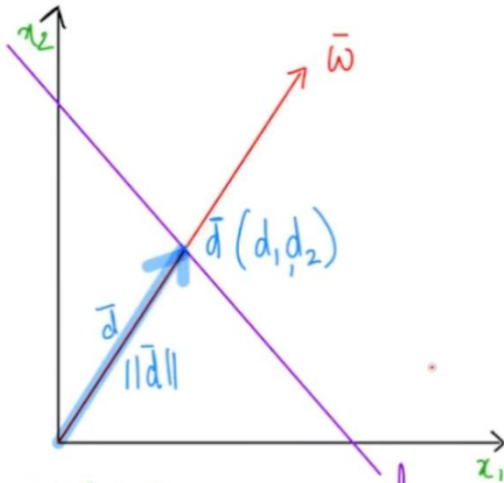
$$(3) \quad \hat{w} = \frac{\bar{w}}{\|\bar{w}\|} \quad (4) \quad \bar{d} = k \hat{w}$$

$k = \text{Constant}$

$$(5) \quad \bar{d} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \Rightarrow \bar{d} = \begin{bmatrix} \frac{k w_1}{\|\bar{w}\|} \\ \frac{k w_2}{\|\bar{w}\|} \end{bmatrix}$$

$$d_1 = \frac{k w_1}{\|\bar{w}\|} \quad d_2 = \frac{k w_2}{\|\bar{w}\|}$$

## Distance Between Origin and a line



$$w_1 x_1 + w_2 x_2 + w_0 = 0 \quad \text{line}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \|\bar{w}\| = \sqrt{w_1^2 + w_2^2}$$

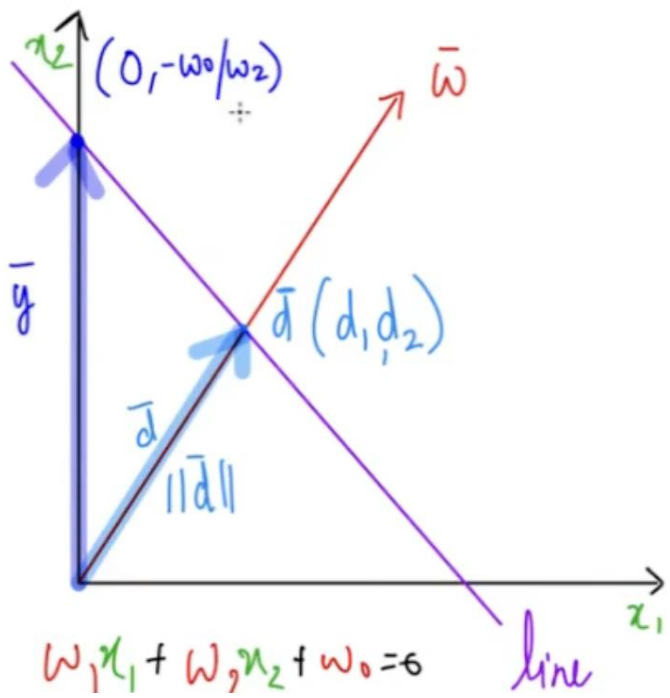
$$(1) \quad \|\bar{d}\| = \sqrt{d_1^2 + d_2^2}$$

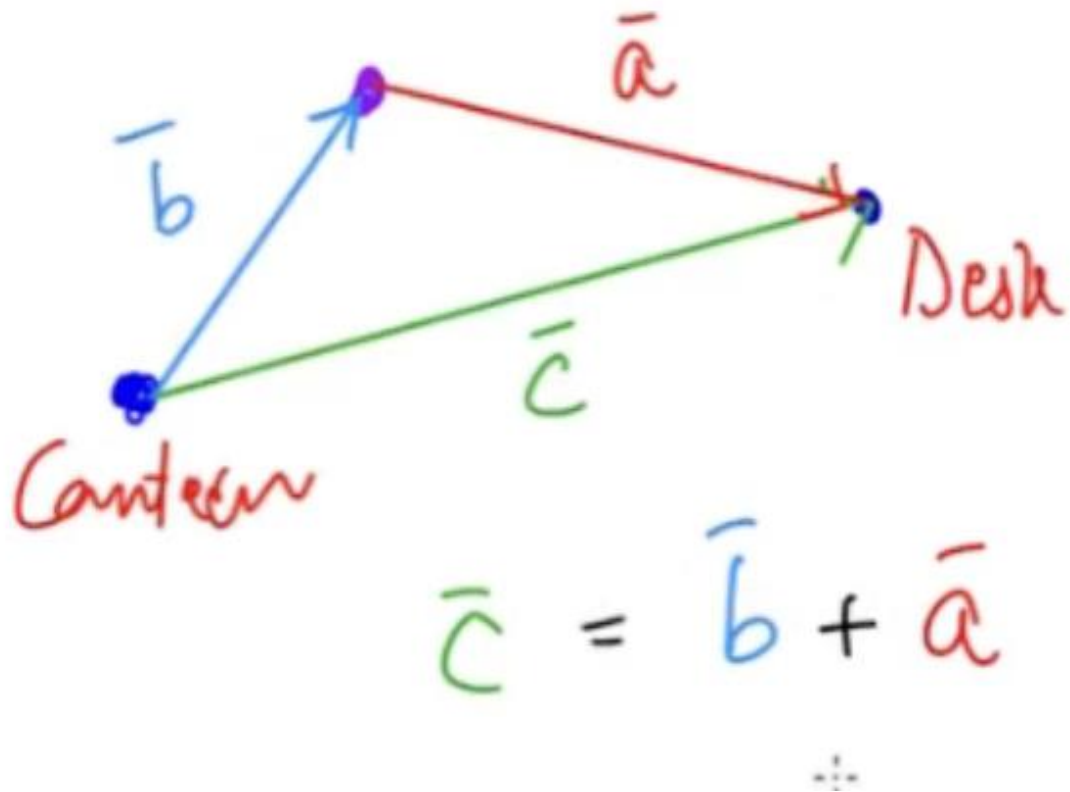
$$(2) \quad w_1 d_1 + w_2 d_2 + w_0 = 0$$

$$(3) \quad \hat{\bar{w}} = \frac{\bar{w}}{\|\bar{w}\|} \quad (4) \quad \bar{d} = k \hat{\bar{w}} \quad \text{Constant}$$

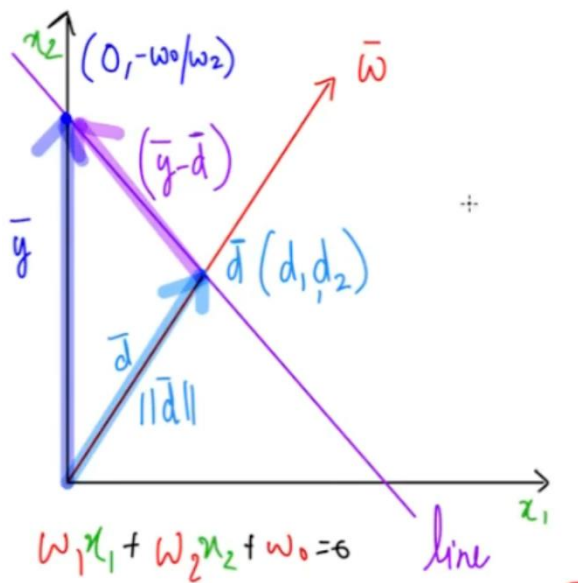
$$(5) \quad \bar{d} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \Rightarrow \bar{d} = \begin{bmatrix} \frac{k w_1}{\|\bar{w}\|} \\ \frac{k w_2}{\|\bar{w}\|} \end{bmatrix}$$

$$d_1 = \frac{k w_1}{\|\bar{w}\|} \quad d_2 = \frac{k w_2}{\|\bar{w}\|}$$





Q



Using the equation of d vector and the value of k we can simplify it as follows

$$\bar{d} = \begin{bmatrix} k \frac{w_1}{\|\bar{w}\|} \\ k \frac{w_2}{\|\bar{w}\|} \end{bmatrix} \Rightarrow \bar{d} = \begin{bmatrix} \frac{w_1}{\cancel{\|\bar{w}\|}} \times \frac{-w_0 \cancel{\|\bar{w}\|}}{(w_1^2 + w_2^2)} \\ \frac{w_2}{\cancel{\|\bar{w}\|}} \times \frac{-w_0 \cancel{\|\bar{w}\|}}{(w_1^2 + w_2^2)} \end{bmatrix}$$

$$\bar{d} = \begin{bmatrix} \frac{-w_1 w_0}{w_1^2 + w_2^2} \\ \frac{-w_2 w_0}{w_1^2 + w_2^2} \end{bmatrix} +$$

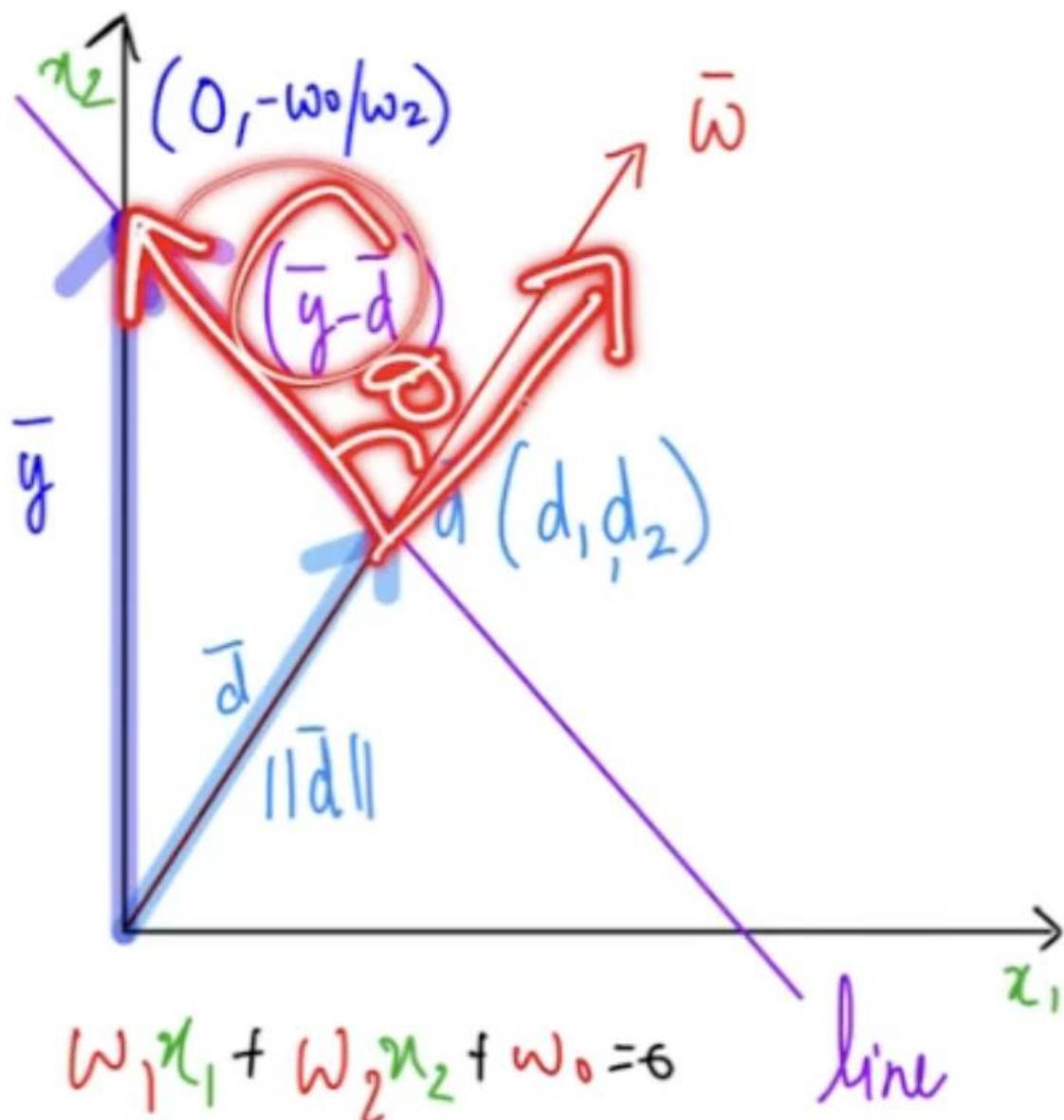
Lets compute the x vector which is vector y subtracted from vector d

$$\bar{y} - \bar{d} = \begin{bmatrix} 0 - \left( \frac{-w_1 w_0}{w_1^2 + w_2^2} \right) \\ \frac{-w_0}{w_2} - \left( \frac{-w_2 w_0}{w_1^2 + w_2^2} \right) \end{bmatrix}$$

$$\bar{y} - \bar{d} = \begin{bmatrix} \frac{w_1 w_0}{w_1^2 + w_2^2} \\ \frac{-w_0 (w_1^2 + w_2^2) + w_2^2 w_0}{w_2 (w_1^2 + w_2^2)} \end{bmatrix}$$

Can we find angle between vector d and vector x ie vector (y-d)?





$$\begin{aligned}
 & \underline{\bar{d}^T (\bar{y} - \bar{d})} \quad \bar{d}^T \quad (\bar{y} - \bar{d}) \\
 & \left[ \frac{-w_1 w_0}{w_1^2 + w_2^2} \quad \frac{-w_2 w_0}{w_1^2 + w_2^2} \right] \left[ \begin{array}{c} \frac{w_1 w_0}{w_1^2 + w_2^2} \\ \frac{-w_0 (w_1^2 + w_2^2) + w_2^2 w_0}{w_2 (w_1^2 + w_2^2)} \end{array} \right] \\
 & \frac{-w_1 w_0}{w_1^2 + w_2^2} \frac{w_1 w_0}{w_1^2 + w_2^2} + \frac{-w_2 w_0}{w_1^2 + w_2^2} \frac{-w_0 (w_1^2 + w_2^2) + w_2^2 w_0}{w_2 (w_1^2 + w_2^2)} \\
 & = \frac{-w_1^2 w_0^2}{(w_1^2 + w_2^2)^2} + \frac{w_0^2 w_1^2 + w_1^2 w_2^2 - w_2^2 w_0^2}{(w_1^2 + w_2^2)^2} \\
 & = \frac{-w_1^2 w_0^2 + w_0^2 w_1^2 + w_1^2 w_2^2 - w_2^2 w_0^2}{(w_1^2 + w_2^2)^2} = 0
 \end{aligned}$$

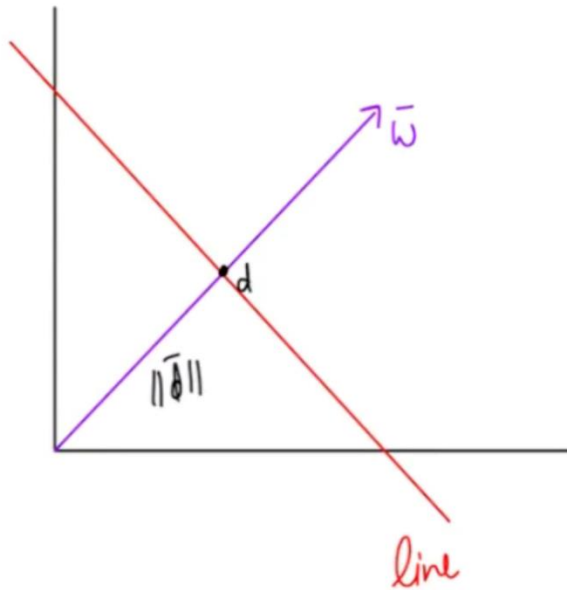
$$\begin{aligned}
 & \bar{d}^T (\bar{y} - \bar{d}) \Rightarrow \cos \theta = 0 \\
 & \cos \theta = \frac{\bar{d}^T (\bar{y} - \bar{d})}{\|\bar{d}\| \|\bar{y} - \bar{d}\|} \Rightarrow \cos \theta = 0 \\
 & \quad \quad \quad \theta = 90^\circ
 \end{aligned}$$

- This is just to understand not asked in interviews
- But from concept understanding perspective it is a very important topic because later on we define any hyperplane using the weight vector also called as the norm, and we'd care less about the lines and only focus on this weight vectors

Now given the line and the weight vector how would you find the distance from origin and line?

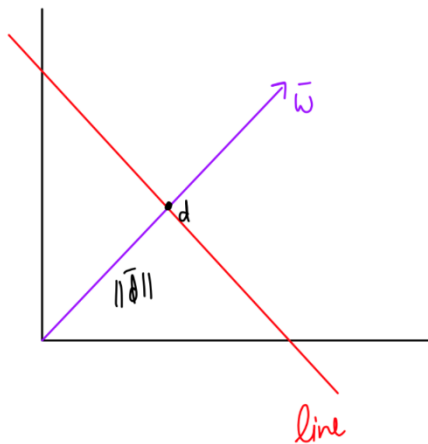
- we need to find  $\|\bar{d}\|$

### Distance between Origin and A line



$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

### Distance between Origin and A line



$$w_1 x_1 + w_2 x_2 + \underbrace{w_0}_{=0} = 0$$

$$\|\bar{d}\| = \frac{w_0}{\|\bar{w}\|}$$

$$Ax + By + C = 0$$

$$d = \frac{C}{\sqrt{A^2 + B^2}}$$

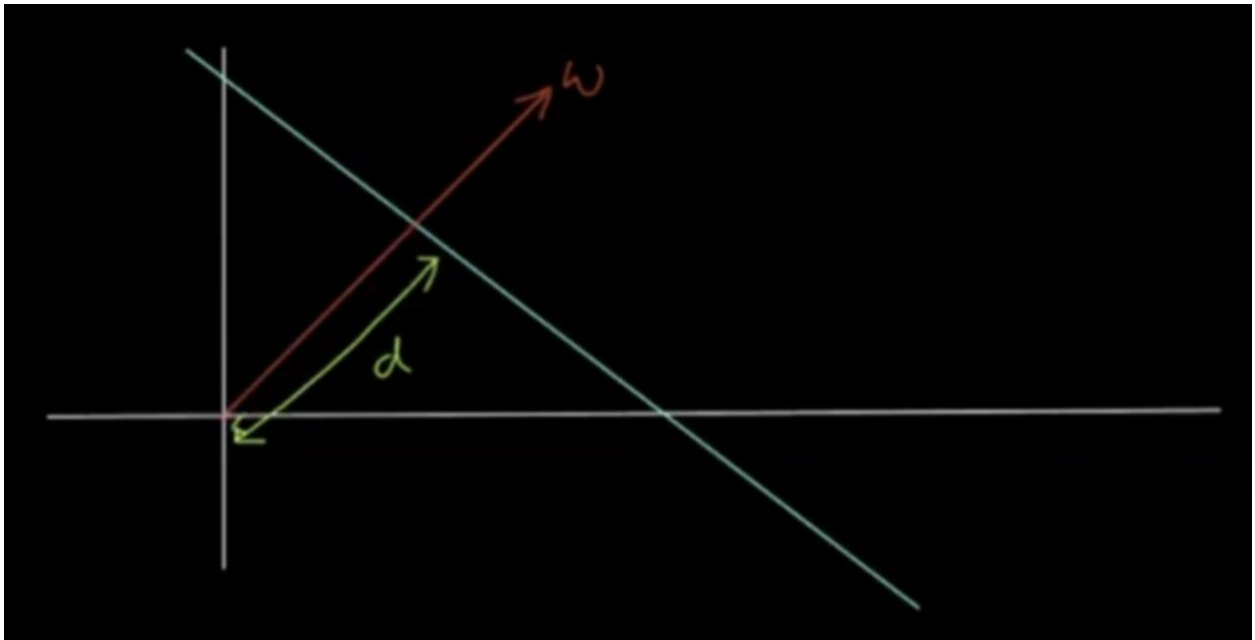
Distance between a line and the origin.

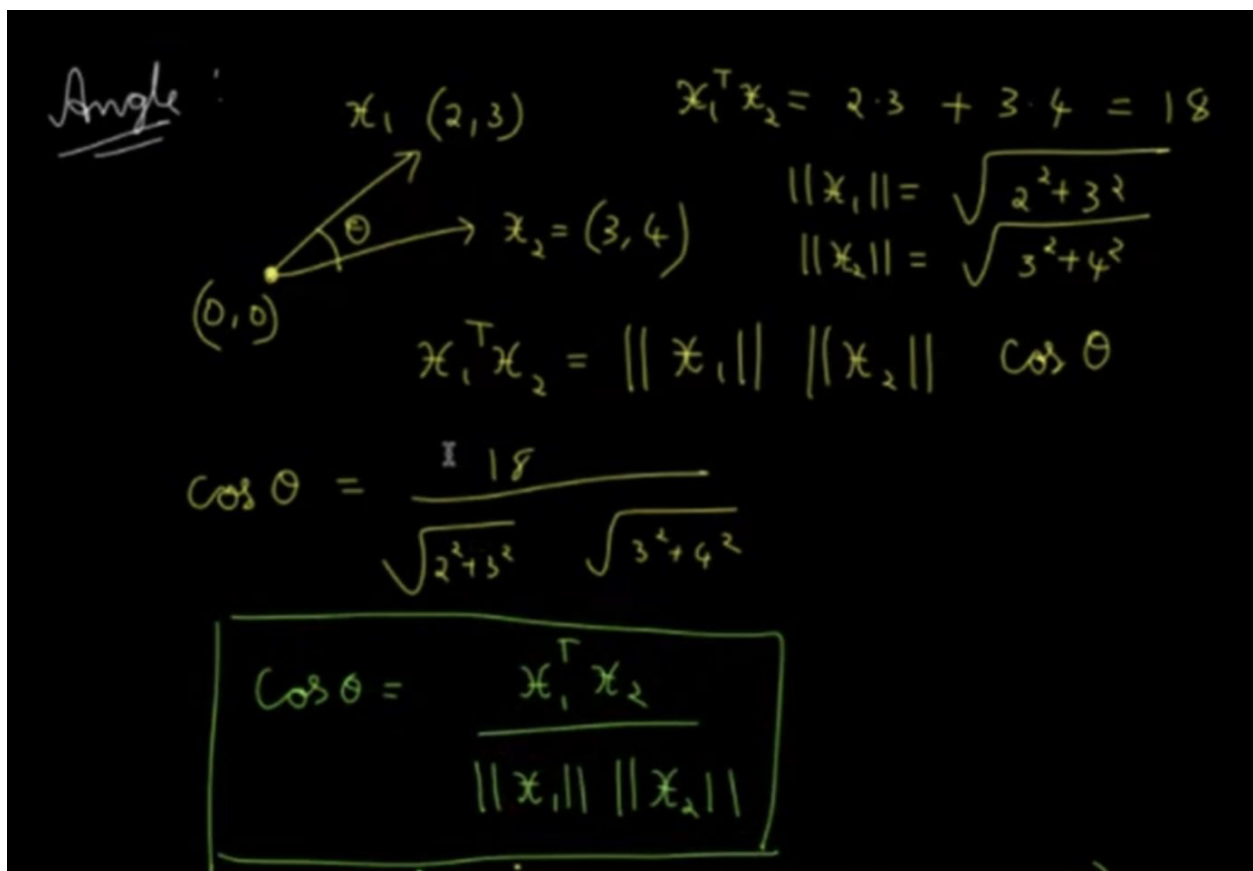
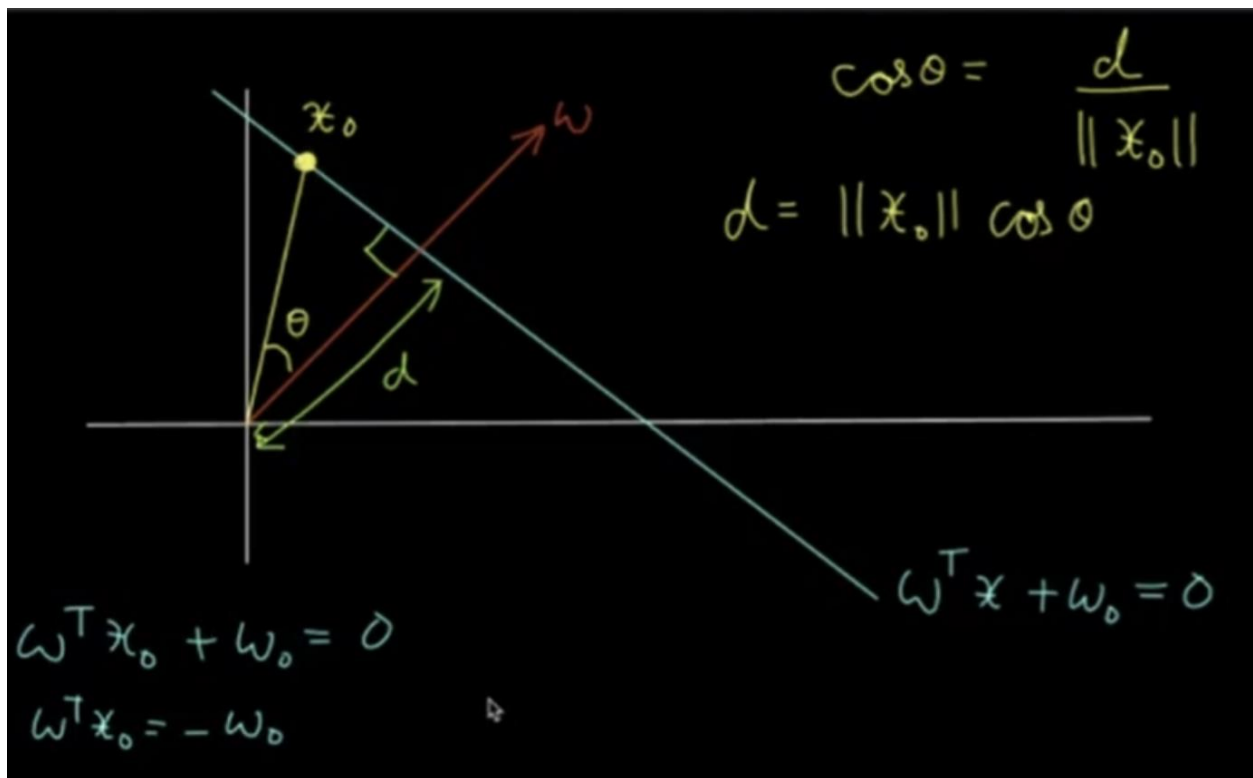
$$\bar{d} = \begin{bmatrix} \frac{-w_0 w_1}{(w_1^2 + w_2^2)} \\ \frac{-w_0 w_2}{(w_1^2 + w_2^2)} \end{bmatrix}$$

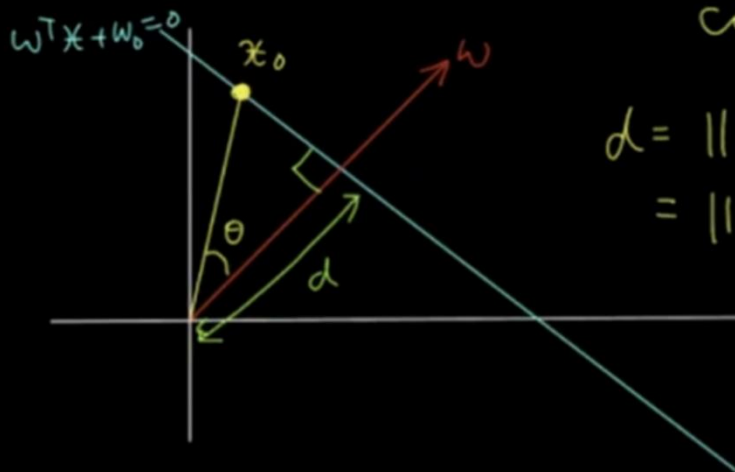
Hence, the distance between the origin and a line  $\bar{w}^T \bar{x} + w_0 = 0$  is given by

$$d = \frac{w_0}{\|\bar{w}\|}$$

$$\begin{aligned} \|\bar{d}\| &= \sqrt{\frac{w_0^2 w_1^2}{(w_1^2 + w_2^2)^2} + \frac{w_0^2 w_2^2}{(w_1^2 + w_2^2)^2}} \\ &= \sqrt{\frac{w_0^2 (w_1^2 + w_2^2)}{(w_1^2 + w_2^2)^2}} = \frac{w_0}{\sqrt{w_1^2 + w_2^2}} = \frac{w_0}{\|\bar{w}\|} \end{aligned}$$







$$\cos \theta = \frac{d}{\|x_0\|}$$

$$d = \|x_0\| \cos \theta$$

$$= \|x_0\| \left( \frac{w^T x_0}{\|w\| \|x_0\|} \right)$$

$$d = \frac{w^T x_0}{\|w\|}$$

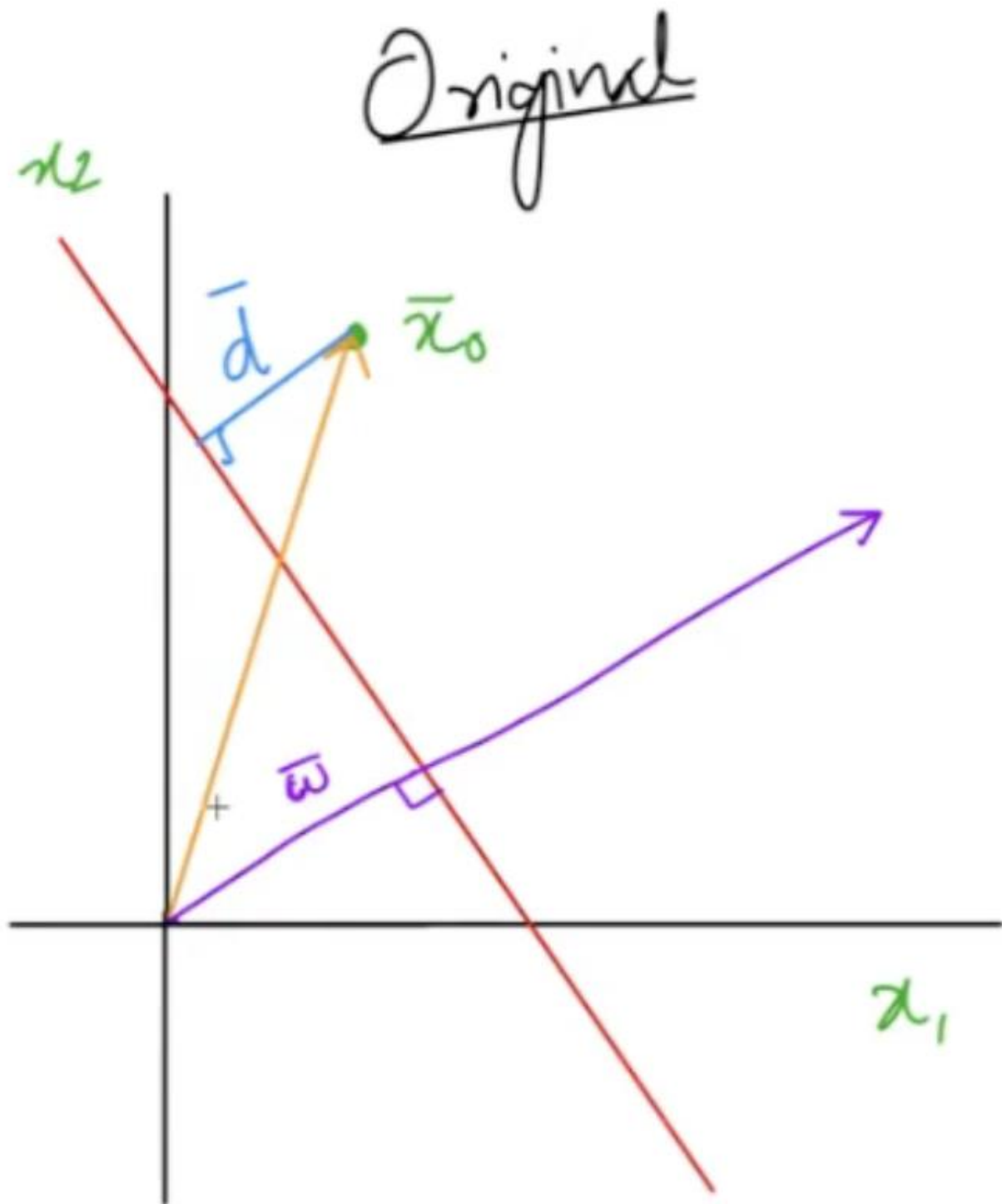
$$= \frac{-w_0}{\|w\|}$$

$$w^T x_0 + w_0 = 0$$

$$w^T x_0 = -w_0$$

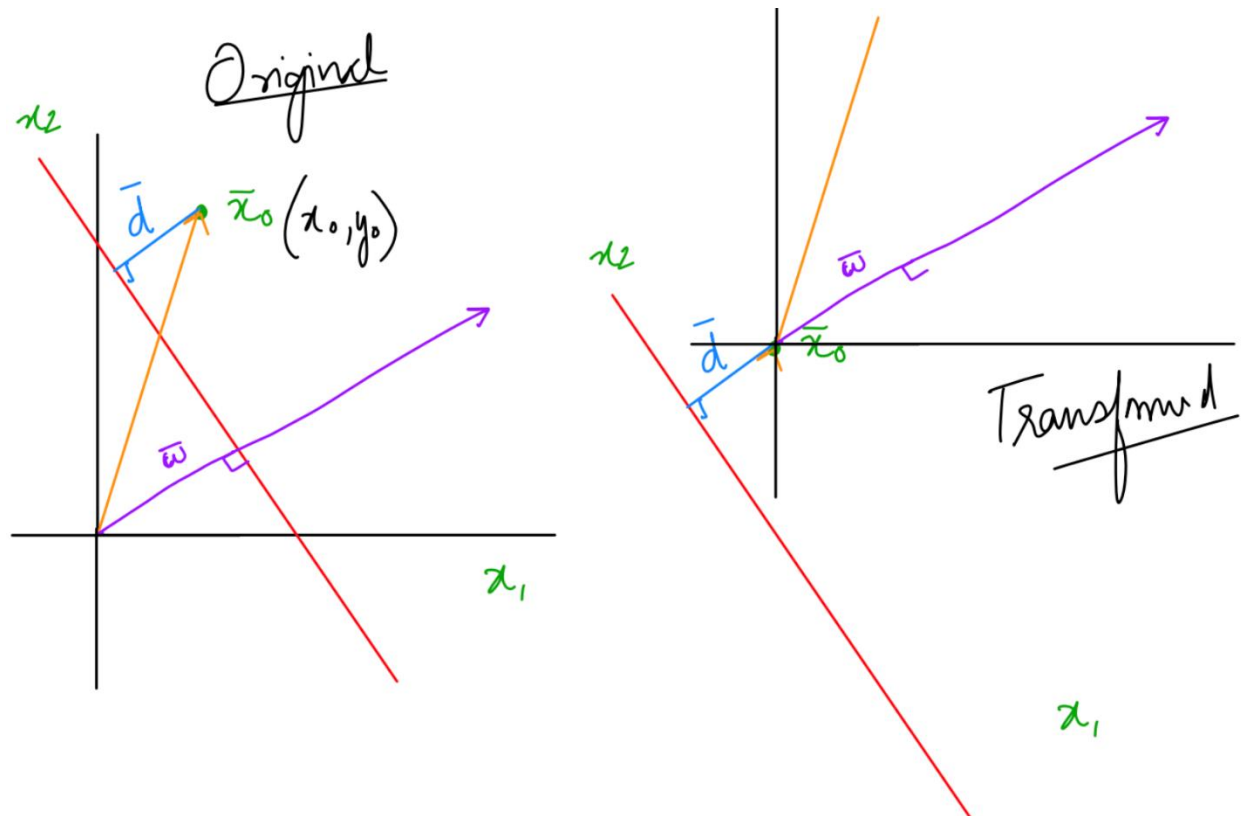
From  $x_0$  and  $w$ ,  
what is  $\cos \theta = \frac{w^T x_0}{\|w\| \|x_0\|}$

SCALER



Now using the shifting concept we studied earlier we can shift the origin to point  $\bar{x}_0$

And now we can see that for the line the  $d$  vector will now act as the weight vector which is perpendicular to our Line.



**Q. How do we represent the equation of line when we shift the line to point?**

- Let say that point  $x_0$  is represented as  $(x_0, y_0)$
- Thus our equation would be  $w_1(x_1 + x_0) + w_2(x_2 + y_0) + w_0 = 0$
- After simplifying the equation and using the distance formula we can compute the distance  $d$



$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$w_1(x_1 + x_0) + w_2(x_2 + y_0) + w_0 = 0$$

$$\underbrace{w_1 x_1 + w_2 x_2}_{w_0'} + w_1 x_0 + w_2 y_0 + w_0 = 0$$

$$d = \frac{w_0'}{\|w\|} = \frac{w_1 x_0 + w_2 y_0 + w_0}{\sqrt{w_1^2 + w_2^2}}$$

$$Ax + By + C = 0$$

$(x_1, y_1)$

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$w_1(x_1 + x_0) + w_2(x_2 + y_0) + w_0 = 0$$

$$w_1 x_1 + w_2 x_2 + w_1 x_0 + w_2 y_0 + w_0 = 0$$

$$d = \frac{w_0'}{\|\bar{w}\|} = \frac{w_1 x_0 + w_2 y_0 + w_0}{\sqrt{w_1^2 + w_2^2}} = \frac{w_0'}{\|\bar{w}\|}$$

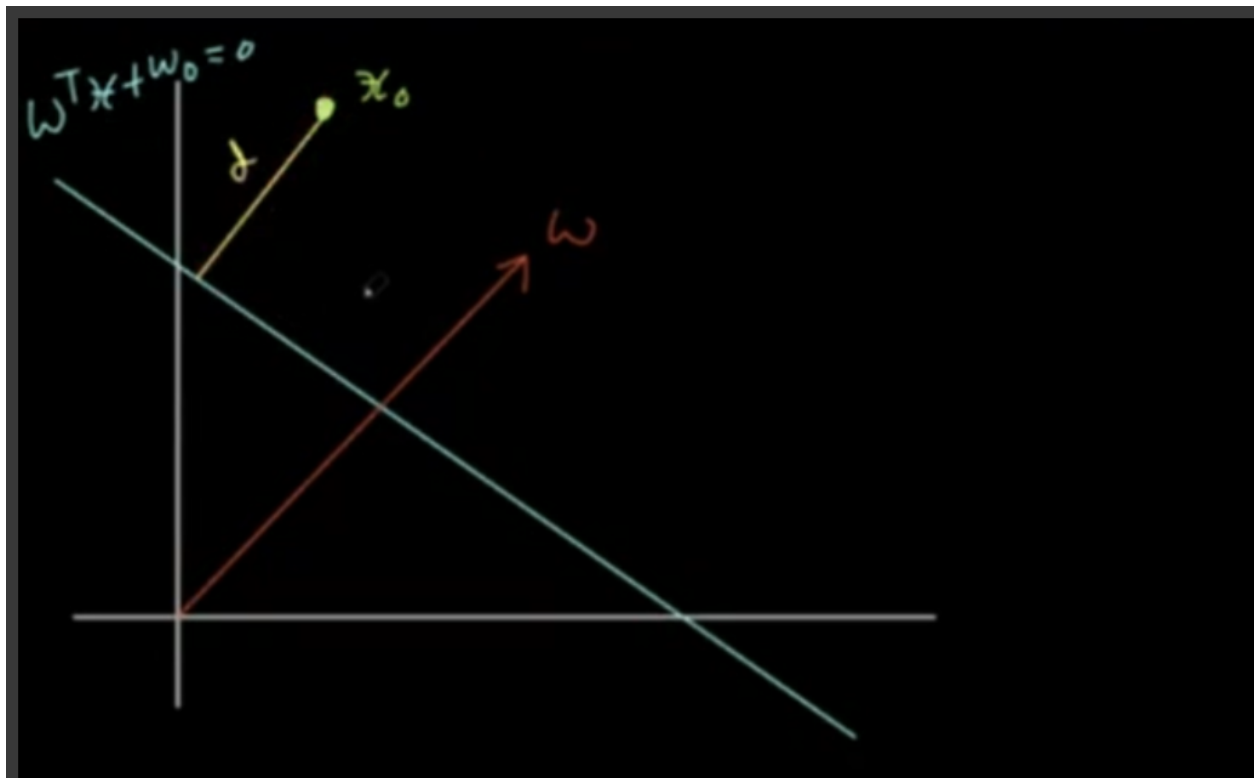
$$Ax + By + C = 0$$

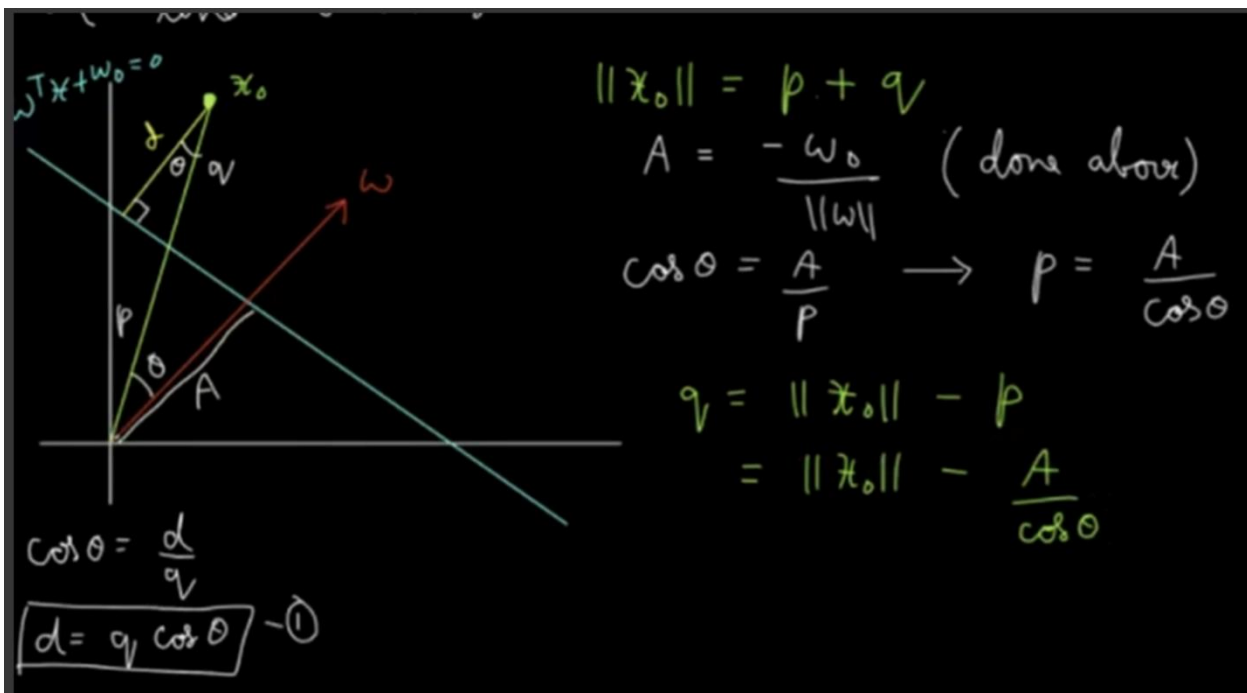
$$(x_1, y_1)$$

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \bar{x}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$





$\cos \theta = \frac{d}{q_v}$   
 $\boxed{d = q_v \cos \theta} \text{---(1)}$

$= \|x_0\| - \frac{A}{\cos \theta} \text{---(2)}$

$d = \left( \|x_0\| - \frac{A}{\cos \theta} \right) \cos \theta$   
 $= \|x_0\| (\cos \theta) - A$   
 $= \cancel{\|x_0\|} \left( \frac{w^T x_0}{\|w\| \cancel{\|x_0\|}} \right) - \left( \frac{-w_0}{\|w\|} \right)$

$\boxed{d = \frac{w^T x_0 + w_0}{\|w\|}}$